This problem set is on a mixture of topics. It will not be marked; no need to hand in your work. However, working on these problems will probably be in your interest. Solutions will be discussed in the tutorials of May 5th/6th.

It is strongly recommended that you try to solve the problems *before* the tutorial. If you are reading the problems for the first time during tutorial, you are unlikely to understand the solutions.

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1. A circular conducting loop with radius R, centered at the origin, rotates around the y-axis, so that the angle $\theta(t)$ between the normal to the loop and the z-axis varies as $\theta(t) = \Omega t$. A constant magnetic field points in the z-direction: $\mathbf{B} = B_0\hat{k}$. Find the EMF induced in the loop.

2. Two metallic (perfectly conducting) solid spheres each have radius R. One of them carries charge $+Q$ and is centered at $(0, d/2, 0)$. Assume $R < d/2$, so that the origin is outside the sphere. The other sphere is centered at $(0, -d/2, 0)$ and carries charge $-Q$.
   (a) What is the electric field inside either of the spheres?
   (b) What is the electric field (magnitude and direction) at a point $(x_0, 0, 0)$ on the x axis?
   (c) What is the electric field (magnitude and direction) at a point $(0, y_0, 0)$ on the y axis, which is outside both of the spheres? Approximate this expression for points far from the dipole arrangement, i.e., for $y_0 \gg d$.
   (d) What are the magnitude and the direction of the electric dipole moment?

3. (a) The electric field in some region is changing with time:
$$\mathbf{E} = -\left(\frac{6K_0 t}{\epsilon_0}\right)\hat{k}$$
   Find the displacement current density $\mathbf{J}_D$.
   (b) If the magnetic field in this region is
$$\mathbf{B} = \mu_0 K_0 \left(3y\hat{\imath} - 3x\hat{\jmath}\right)$$
   then find the current density $\mathbf{J}$.

4. Three points P, Q and R have coordinates

$$P: \quad (l_0, 5l_0, 0) \qquad\qquad Q: \quad (4l_0, 5l_0, 0) \qquad\qquad R: \quad (l_0, 0, 0)$$

   (a) The electric field is given to be uniform: $\mathbf{E} = (7\ W/l_0)\hat{\imath}$. Find the potential difference V_{PQ} between points P and Q, and the potential difference V_{PR} between points P and R.

   Hint: a sketch showing a top view of the x-y plane might help.

   (b) Now consider the electric field to be given by

$$\mathbf{E} = (2x\ W/l_0^2)\hat{\imath}$$

   Find the potential difference V_{PQ} between points P and Q, and the potential difference V_{PR} between points P and R.

5. An infinite wire carrying current I runs along the y axis; the current flows from $y = -\infty$ to $y = +\infty$ through the origin.

   A square loop of wire lies on the xy plane, with the four corners having coordinates (x_0, y_0), $(x_0 + L, y_0)$, $(x_0 + L, y_0 + L)$, and $(x_0, y_0 + L)$.

   A sketch showing a top view of the xy plane might help.

   (a) Find the magnetic flux through the square loop. The magnetic field is created by the current through the long wire.

   (b) Imagine that the square loop moves away from the long wire with speed v, so that $x_0(t) = vt$ but y_0 and L are constant. Find the EMF generated in the square loop.

   (c) Imagine instead that the square loop moves in the y-direction (parallel to the long wire) with speed v, so that $y_0(t) = vt$ but x_0 and L are constant. Explain why there is no EMF generated in this situation.